

ABSTRACT ALGEBRA 1 - HOMEWORK 5- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

- (1) **Problem 1*:** Let $(G, \star, e)$ be a group. We define the center of G to be the subset:

$$(1.1) \quad Z(G) = \{g \in G \mid \forall h \in G \ h \cdot g = g \cdot h\}$$

Prove that $Z(G)$ is a subgroup.

Consider the alternating group A_n on n -letters. Which is the center of A_n ?

- (2) **Problem 2**:** Let $(G, \star, e)$ be a group and S be a set. Let $\phi : G \mapsto \Sigma(S)$ be a group homomorphism. Prove that the following map of sets:

$$\begin{aligned} \mu : G \times S &\mapsto S \\ (g, s) &\mapsto \phi(g)(s) \end{aligned}$$

is an action.

- (3) **Problem 3**:** Define an action of Sym_3 on $\mathbb{R}^3$ as follows: for $\sigma \in Sym_3$ and $v = (x_1, x_2, x_3) \in \mathbb{R}^3$, set

$$\sigma(v) = (x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}).$$

- (a) Show that the subspace $V = \{(x_1, x_2, x_3) \mid x_1 + x_2 + x_3 = 0\}$ has the following property: if $v \in V$, then $\sigma(v) \in V$.
 (b) Can you find a line (one-dimensional subspace) which has this same property?